

lec 13

Introduction to computing

Fibonacci numbers:-

↳ sequence

It is used focus
on the object like
in cameras.

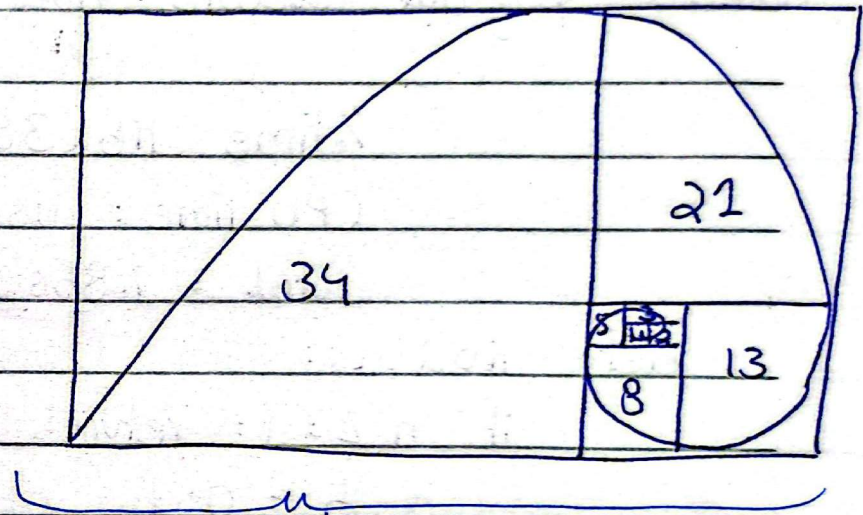
```
def fib(n):
```

```
    if  $n \leq 1$ :
```

```
        return n
```

```
    else:
```

```
        return fib(n-1) + fib(n-2)
```



called golden spiral.

Agar ham na time dealna hu kitnay time ma hamara program execute hua ha.

%time fib(35)

CPU time : user 7.66s , sys : 205ms,
total : 7.86s wall time : 14.2s

```
def fib2(n):
```

```
    if n <= 1: return n
```

```
    a, b = 0, 1
```

```
    for i in range(1, n+1):
```

```
        a, b = b, a+b
```

```
    <-
```

```
        n = a+b
```

```
        a = b
```

```
        b = n
```

```
    return a
```

alternative
method

3

reading assignment:

Style guide for python

<https://www.python.org/dev/peps/pep-0008/>

Sequences :-

programming

fundamentals

→ data storage

→ Branching statements

→ Iterations

- A sequence is an ordered collection of values.
- A powerful, fundamental abstraction.
- Not instances of a particular built-in type.
- Many types but they all share common behaviour:

- length

- element selection

(پیش کیایا جاتا ہے)

Composing programs.

② type of sequence

Lists:-

list are a sequence that have arbitrary length. Extremely powerful and useful.

Social media and feed are in list
the list is very useful in real world

List in python:-

```
digits = [1, 8, 2, 8, 4]
```

```
[1, 8, 2, 8, 4]
```

no of $\leftarrow \text{len}(\text{digits})$

items

5

members

elements
in this list

$\text{len}()$

$\hookrightarrow$ tells us the total
elements of list

For new lang learning:-
1) case sensitivity

2) In matlab the index is 1-based index.

position
digits [0] → to abstract 1 element # this is called 0 base index
index

1 in digits
True ↓
boolean operator

it return true or false

digits = [1, 2, 3, 4, 5]
digits [0:2] # stop
[1, 2]
2

digits [2:]
[4, 5] [4, 6, 7, 9]

digits [1, 5, 4, 6, 7, 9]

↓ ↓ ↓
... -3, -2, -1 index

digits [fun()]
↓
I also do that thing as well.

In list order is important

digit [0:2] imp: 0 ki left boundary se start karo or 2 ki left boundary se pehlay ruk jao